MAYANK SHARMA

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Research interests: latent reasoning, world models (state tracking), reliable ML evaluation, and scientific machine learning.

EDUCATION

Indian Institute of Technology Jodhpur Aug 2023 – May 2027 (expected)
B.Tech. in Engineering Science | Majors in Computer Science and Electrical Engineering
Selected coursework: Machine Learning, Deep Learning, Computer Vision, Probability and Stochastic Processes, Algorithms, Operating Systems, Signals and Systems, Scientific Computation.

PUBLICATIONS AND MANUSCRIPTS

Accepted

[1] **M. Sharma**, A. Kamboj. **UnitScope: Limits and Conditions for Bit-Exact Unit Transport in Stateful Safe RL.**
NewInML Workshop at NeurIPS 2026 | Accepted poster, 2026 (non-archival workshop) | OpenReview

[2] **M. Sharma**, R. K. Mourya, S. Srivastava. **Physics-Induced Compressibility in CNN-Aided Zak-OTFS Channel Estimation: A Compact Estimator and Its Limits.**
IEEE ANTS 2026 | Accepted, 2026

[3] **M. Sharma**, R. K. Mourya. **MacCode Lab: A Reproducible Single-Machine Harness for Execution-Guided Code Agents, and When Repair Actually Helps.**
Agentic AI Evaluation Workshop at ACM KDD 2026 | Accepted workshop paper, 2026 | Workshop

Under Review

[4] **M. Sharma**, R. K. Mourya, P. Mazumder. **Logit-Origin Centering for Singleton Test-Time Adaptation.**
Neurocomputing | Under journal review, 2026 | arXiv:2608.01074

[5] **M. Sharma**, R. K. Mourya. **Balanced Prototype Codes Beyond the Simplex Barrier for Open-Set Recognition.**
AAAI 2027 | Under review; earlier preprint on arXiv | arXiv:2606.01883

Preprints

[6] **M. Sharma**, R. K. Mourya. **Structural Hawkes Ambiguity in Near-Critical Market Making.**
SSRN 6777719 | Preprint, 2026 | Zenodo

[7] **M. Sharma**, M. Baruah. **Leakage-Safe Financial News Sentiment Trading in United States Equities.**
SSRN 6813859 | Preprint, 2026 | SSRN

RESEARCH EXPERIENCE

Shared randomness and sampler implementations Sep 2026 – Present
Independent research | Manuscript in development

- Studying when exact samplers need memory to optimize paired evaluations under shared randomness. Built exact-arithmetic proof checks and a held-out scheduling benchmark that exposed limits of the proposed compiler.

Latent reasoning and mechanistic evaluation Aug 2026 – Present
B.Tech. project, IIT Jodhpur | Supervisor: Prof. Deeksha Varshney

- Studying how positional spacing, attention routing, and answer format affect the execution of latent reasoning on synthetic state-tracking tasks.
- Tested a format that completes the final state update before answering. On a 0.87M-parameter model, held-out length-64 accuracy was 99.8% and 99.3% across two seeds; an extra-step control gave mixed results.
- Extended the experiments to Qwen2.5-0.5B with LoRA and a task-specific vocabulary. In one seed, wide-gap training reached 97.3% accuracy at length 64 versus 2.5% with narrow gaps, after training only on lengths 1–7.

UnitScope and safe reinforcement learning Aug 2026 – Present
Independent research with Abhinav Kamboj

- Derived conditions for bit-exact power-of-two unit rescaling under a fixed execution graph and safe numerical range. Tested repaired FSRL and OmniSafe adapters on PointGoal at scale 2 for 60,000 post-switch steps per branch, with two seeds per library on a pinned CPU setup.
- Ran preregistered local and closed-loop audits, showing why finite-tolerance update certificates alone do not establish equivalent training behavior.

Full-waveform inversion and uncertainty quantification May 2026 – Present
Research collaboration | Stanford × SLB

- Evaluated flow-based seismic inversion under geology and acquisition shifts; tested physics-guided input adaptation and identified failure under narrow-band measurements.
- Reviewed flow matching, diffusion, and simulation-based inference for uncertainty-aware inversion; prepared a 45-minute group briefing on calibration and physics-consistency checks.

ML evaluation and efficient implementations Jan 2026 – Present
Independent research, IIT Jodhpur

- Built MacCode Lab, a reproducible harness for execution-guided code agents on Apple Silicon; evaluated when repair improves hidden-test performance and when public tests mislead.
- Studied balanced prototype geometry for open-set recognition and evaluated its behavior across multiple scoring rules.
- Implemented matrix-free Zak-OTFS operators that reduced saved-benchmark equalizer latency from 7.67 to 2.18 ms; prepared a reproducible release of the accepted ANTS estimator.
- Developed llama.cpp Metal kernels on Apple Silicon (flash-attention prefill, grouped-query decode, Q8_0 matmul), giving a local 27B model 2.4× faster 128k-token prefill and 20–60% faster generation with identical greedy outputs.

Singleton test-time adaptation Dec 2025 – Present
IIT Jodhpur | Advisor: Prof. Pratik Mazumder

- Proposed PLOC, a label-free correction for streaming tabular classification that keeps the source model frozen and stores one running scalar per logit.
- Evaluated five benchmarks, three architectures, and five source checkpoints; developed a deferred variant that preserves the source ranking and AUROC exactly.

Financial machine learning May 2025 – Jul 2026
University of Birmingham, Dubai Campus | Advisor: Prof. Monita Baruah

- Developed news-sentiment trading pipelines over 33 US equities using out-of-fold predictions, embargoed time splits, and transaction-cost-aware backtests.

INDUSTRY EXPERIENCE

Quantitative Data Engineering Intern May 2025 – Jul 2025
Detachmint Ventures Pvt. Ltd. | Hybrid, India

- Built Python/Pandas pipelines for 80+ parallel one-minute data streams and evaluated liquidity, seasonality, and volatility filters out of sample.

BRAIN Research Consultant Dec 2023 – Sep 2025
WorldQuant | Remote

- Researched and backtested 2,000+ equity alpha candidates over ten years of data, with attention to out-of-sample stability, turnover, and signal decay.

SELECTED HONORS

- **STEMS Residential Camp, Chennai Mathematical Institute (2026):** selected among 40 students nationwide for a fully funded mathematics camp.
- **ICPC 2025 Online Winter Challenge (Huawei):** Global Rank 192.
- **Baruch Pre-MFE Linear Algebra Program:** selected among 40 worldwide; earned Distinction.
- **IIT Roorkee Trading Heist 2025:** 6th of 631 teams.
- **Codeforces Expert:** peak rating 1736.

TECHNICAL SKILLS

Languages	Python, C++, SQL, Bash
ML frameworks	PyTorch, MLX, scikit-learn, XGBoost
Scientific stack	NumPy, SciPy, pandas, statsmodels
Tools	Git, Linux, Jupyter, LaTeX
Methods	Experiment design and Apple Silicon profiling